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The Effect of Attentional Locus on Multisensory Congruence: An ERP Study

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ABSTRACT

Extensive research has established the pivotal role of the congruence effect for multisensory integration of audiovisual stimuli. However, little is known about whether the attentional locus on one or other sensory domain has a differential impact on the neurophysiological dynamics of the congruence effect, particularly under conditions of frequent incongruence. Therefore, the aim of this study was to compare the temporal and topographic features of event-related potentials (ERPs) associated with the congruence effect between attended sensory modalities. Thirty healthy young adults underwent a crossmodal (visual/auditory) attention task with simultaneous audiovisual stimuli (75% incongruent trials, 25% congruent trials) during an EEG recording. Behaviorally, a stronger congruence effect for visual attention than for auditory attention was observed. The congruence effect ERP (ERPce) was extracted by subtracting the ERPs of incongruent trials from the ones of congruent trials for each attentional locus modality (visual/auditory). The ERPce mainly consisted of an N1-P2-P3 complex with a left fronto-central distribution and similar amplitudes regardless of the attended sensory modality, which was associated with behavioral performance. Interestingly, the attentional locus significantly influenced the temporality of the congruence effect, as revealed by delayed latencies of the ERPce for auditory versus visual attention. These results suggest that the attentional locus plays a significant role in integrating simultaneous congruent (though irrelevant) signals with relevant ones in a context of broad incongruency.

1 | Introduction

Since our perceptual capacity is limited, fixed, and automatic, we cannot process all the stimuli overloading our senses. To address this, we rely on attentional control mechanisms which allow us to direct and prioritize brain resources toward a relevant subset of such information in line with our goals, while blocking the processing of irrelevant information coming from other sensory modalities in the case of crossmodal situations (Buschman and Kastner 2015). However, frequently we encounter situations in everyday life when the unattended sensory modality (e.g., visual) is congruent with the attended one (e.g., auditory). For example, in a noisy restaurant, while trying to listen to someone at our table, the movements of her lips in congruence with her speech may help us understand the message, even though those movements are not the primary focus of attention. In those cases, the processing of the irrelevant sensory modality drives and enhances the one of relevant information, producing a congruence effect (Solís-Vivanco et al. 2021). The ability to detect and integrate congruent signals from the sensory modality

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outside the primary focus of attention, in parallel with attended ones, represents a core feature of multisensory integration and can strongly modulate perceptual and attentional performance (Busse et al. 2005).

From a neurophysiological perspective, the event-related potential (ERP) evidence of multisensory integration of audiovisual (AV) signals has been reported mainly through the N1 and P2 components, which are thought to play distinct functional roles in the processing of multisensory information (Klucharev et al. 2003). For instance, the N1 component has been associated with the integration of spatially and temporally congruent AV signals (Paris et al. 2017; Vroomen and Stekelenburg 2010). Previous research has shown that when visual information predicts the onset of congruent auditory input, there is a reduction of visual P50 amplitudes and a reduction and acceleration of the auditory N1 response, relative to incongruent stimuli (Baart et al. 2014; Stekelenburg and Vroomen 2012). This suggests that congruent visual cues can facilitate early auditory processing. In contrast, other reports have shown an increase in N1 amplitude in response to congruent AV stimuli compared to incongruent ones, particularly under conditions of simultaneous presentation, suggesting that congruence may enhance sensory gain rather than predictability per se (Ho et al. 2015). Regarding the P2 component, a decrease in its amplitude has been interpreted as a marker of the integration of more complex stimulus features, such as phonetic or semantic content (Stekelenburg and Vroomen 2007). For example, Kokinous et al. (2015) reported reduced P2 amplitudes for emotionally congruent face–voice pairs compared to incongruent ones, and Baart et al. (2014) found lower P2 amplitudes in response to phonemically congruent AV syllables compared to incongruent ones.

While these findings indicate that congruence between sensory modalities modulates both the timing and magnitude of early ERP components, we note that results based on simultaneously presented AV stimuli diverge from those using temporally asynchronous presentations. This discrepancy highlights the need to distinguish between facilitation due to temporal predictability and facilitation driven by multisensory congruence under simultaneous input.

The multisensory integration of AV stimuli is influenced not only by spatial, temporal, and phonetic congruence within them, but also by the proportion of incongruent stimuli relative to congruent ones (Atalay and Misirlisoy 2012; Grandjean et al. 2013; Schmidt 2013; Schmidt and Besner 2008). For instance, when the proportion of incongruent/congruent stimuli is oddball-type (75/25%), the attentional system adapts by reallocating resources toward relevant information, leading to an anticipatory blocking of sensory processing of irrelevant stimuli (Lowe and Mitterer 1982; Schmidt et al. 2014; Solis-Vivanco et al. 2018). In these cases, the brain's ability to integrate multisensory congruent signals effectively can be reduced, thereby decreasing the congruence effect (Sarmiento et al. 2012).

In addition to the effects of the proportion of incongruent information, there is considerable evidence from behavioral and neurophysiological studies indicating that attention can be biased in crossmodal tasks (Chen and Spence 2010; Roelofs 2005; Shimada 1990; Yuval-Greenberg and Deouell 2009). For instance, visual stimuli capture attention more effectively than auditory ones when the AV signals are presented simultaneously or closely in time (Bernstein et al. 2008; Besle et al. 2009; Klucharev et al. 2003; Molholm et al. 2004; Pilling 2009; Sinnett et al. 2007; Stekelenburg and Vroomen 2007; Stuart and Carrasco 1993; van Wassenhove et al. 2005; Yao et al. 2018; Yuval-Greenberg and Deouell 2009). Therefore, visual stimuli (even though irrelevant) may be automatically processed in a biased way, leading to an advantage in multisensory processing with respect to auditory signals. On the contrary, whether voluntary allocation to auditory information might impact the congruence effect with visual information remains to be explored.

In summary, understanding whether attentional allocation to one or another sensory modality influences the N1-P2 components during the processing of simultaneous and congruent AV signals—especially under conditions of frequent incongruence—is crucial to identify how the brain integrates congruent but task-irrelevant AV information in crossmodal attention contexts. However, to date, no study has comprehensively addressed (1) the modulation of N1-P2 components as a function of AV congruence, (2) under simultaneous rather than predictive presentation of stimuli, and (3) while manipulating the locus of attention. Given these gaps in the literature, the present study was exploratory in nature. We aimed to examine how simultaneously presented congruent and incongruent AV stimuli are processed during a crossmodal attention task. Specifically, we explored whether the congruence effect—defined here as the difference of ERP (N1 and P2) amplitudes and latencies, and behavioral responses (accuracy and reaction times), between congruent and incongruent trials—varies as a function of the attended sensory (visual or auditory) modality. Additionally, we explored the extent to which neural and behavioral congruence effects are related. This approach allowed us to investigate how crossmodal attention modulates early AV integration processes in the absence of temporal predictability.

2 | Methods

2.1 | Participants

Thirty healthy Mexican participants (15 females; mean age 23.19 ± 3.54 years; age range 19 and 30 years) were included in the study. Participants were recruited from the National Autonomous University of Mexico (UNAM) in Mexico City. All participants had no history of psychiatric, neurological, or drug abuse/dependence disorders. Participants with a global severity index higher than 1.2 from the Symptom Check List 90 (SCL-90) (Derogatis et al. 2008) or a Montreal Cognitive Assessment (MoCA) score < 24 points were excluded from the study (Milani et al. 2018). All participants had normal or corrected vision and reported normal hearing, which was confirmed by achieving at least 90% accuracy in unimodal attention tasks (visual and auditory, 48 trials each) administered prior to the crossmodal task.

A priori power analysis was conducted using G*Power software (Faul et al. 2007) to determine the target sample size. We used the parameters for a repeated-measures ANOVA with one group and four conditions (visual-congruent, visual-incongruent, auditory-congruent, auditory-incongruent), specifying a

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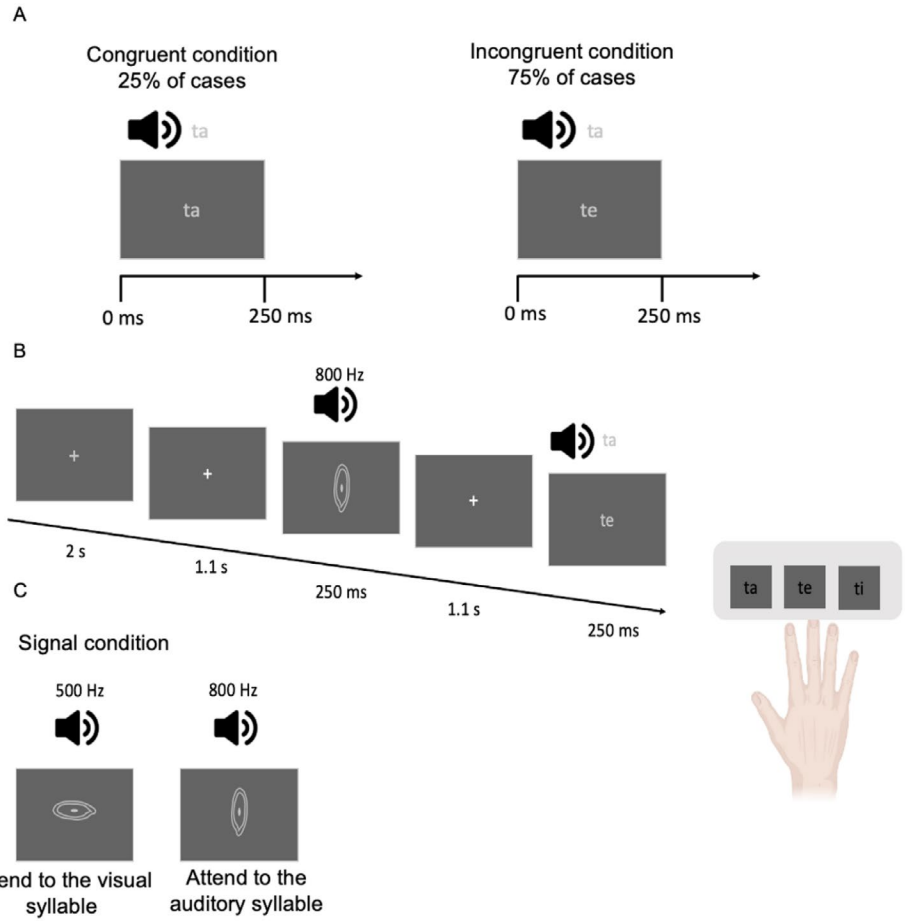


FIGURE 1 | Experimental conditions of the modified crossmodal attention paradigm with (A) congruent and incongruent audiovisual stimuli. (B) In each trial, a bimodal cue (auditory/visual) instructed the participants on the sensory modality (visual or auditory) they should attend to, while ignoring simultaneous information from the other modality. (C) Examples of the cues used in each trial to indicate the relevant sensory modality.

medium effect size ($f = 0.30$), an alpha level of 0.05, and a desired power of 0.95. The analysis indicated that a minimum sample size of 26 participants would be needed to detect the expected effects.

The study was approved by the local Review Board and Ethics Committee of the National Institute of Neurology and Neurosurgery Manuel Velasco Suárez and was in accordance with the Declaration of Helsinki. Before any procedure, participants signed an informed consent letter, where the study was clearly explained, and confidential treatment of their data was guaranteed.

2.2 | Experimental Paradigm

Using custom MATLAB scripts (Mathworks Inc.) and the STIM2 software (Neuroscan Inc.), a crossmodal attention paradigm based on the one reported by Solis-Vivanco et al. (2018) was designed. Each trial (~5 s) started with a dark gray background (RGB code 100/100/100) on a screen and a light gray fixation cross (RGB code 200/200/200) at its center for 2 s. Participants could blink or move their eyes during this period. Subsequently, a white fixation cross appeared, followed by a visual cue (same image evoking an eye or rotated 90° to evoke an ear) together with an auditory cue (pure tone of 500 Hz for visual attention or 1000 Hz for auditory attention) that lasted 250 ms each, indicating the relevant sensory modality to attend to during that specific trial (50% to visual stimuli and 50% to auditory). After a post-signal interval of 1100 ms, a pair of visual and auditory stimuli were presented simultaneously. Both the visual and auditory stimuli always consisted of one of three syllables formed by a consonant and a vowel ("ta", "te", or "ti"). Participants were instructed to indicate the syllable they saw/heard (depending on the assignment given by the previous cues in that trial) by pressing one of three possible buttons with their index, middle, or ring finger on a response pad (Figure 1), as quickly and accurately as possible. In 75% of the trials, the stimuli were incongruent (e.g., visual "ta" and auditory "ti"), whilst the remaining trials were congruent (e.g., "te" in both modalities). Visual syllables appeared in light gray (RGB code 200/200/200) in the center of the screen with a dark gray background (RGB code 100/100/100) in Arial font in a size of 4.8°. Auditory syllables were generated using a male voice and were presented binaurally through earphones inserted in the ears at a volume of 70 dB. Their duration was carefully controlled so that all three of them lasted for 250 ms. Syllables were presented with equal probability in both sensory modalities, distributed over 6 blocks, totaling 288 trials. Participants had 2 s to respond. Incongruent trials where participants responded to the distractor syllable were considered interference errors, those where they responded to the third

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option were considered ambiguous errors, and those with no response were considered misses. Reaction times (RT) and accuracy rates were recorded throughout the experiment.

Each trial was 4.7 s long, and the entire experiment lasted for 2 h. Five 3-min breaks were granted along the experimental task.

2.3 | Procedure

The experiment was conducted in a single session for all participants. During the session, inclusion criteria were confirmed, and general information and detailed instructions about the experiment were presented. The participants performed the experimental tasks while sitting on a chair in a dark EEG room and with a distance of 60 cm between their head and the monitor.

2.4 | Behavioral Analysis

We computed the congruence effect (ce) by subtracting the mean reaction times (RT) of the incongruent condition from those of the congruent condition (RTce). The same procedure was applied to analyze the hit rate of the congruence effect (HRce).

2.5 | EEG and ERP Analysis

During the task, an EEG was recorded [SynAmps RT 64-channel Amplifier (Neuroscan Inc.)] with a bandwidth of 0.1–500 Hz and a sampling rate of 1000 Hz from 58 electrodes placed according to the 10–20 International System assembled on an elastic Easycap (Fp1, Fpz, Fp2, AF7, AF3, AFz, AF4, AF8, F7, F5, F3, F1, Fz, F2, F4, F6, F8, FT7, FC5, FC3, FC1, FCz, FC2, FC4, FC6, FT8, C5, C3, C1, Cz, C2, C4, C6, CP5, CP3, CP1, CP2, CP4, CP6, P5, P3, P1, Pz, P2, P4, P6, P8, PO7, PO3, POz, PO4, PO8, O1, Oz, O2). All electrodes were referenced to the right earlobe and off-line re-referenced to the average of both earlobes (A1 and A2). The ground electrode was placed at CPz. Eye movements from the outer canthus and suborbital of the right eye were recorded. During recording, impedances were kept below 5 KΩ, and a 60 Hz notch filter was applied. Curry 7.0 software (Neuroscan Inc.) was used for EEG recording.

To reduce the effects of fatigue, experiments were conducted between 9 am and 1 pm. All data were analyzed using custom Matlab scripts and the Fieldtrip toolbox (Oostenveld et al. 2011). Data were resampled to 250 Hz. EEG segments from −200 to 1000 ms relative to the presentation of the stimuli were obtained. Visual inspection was performed to eliminate trials containing artifacts such as eye movements, blinks, or muscle activity.

After applying a linear detrending and baseline correction (−200 ms to 0 ms relative to the presentation of stimuli) to the selected epochs, averaged ERPs for each experimental condition (congruent/incongruent and visual/auditory) were obtained separately for each participant. An ERP of the congruence effect (ERPce) was obtained for each sensory modality by subtracting the ERPs of incongruent trials from those of congruent trials in each participant.

2.6 | Statistical Analysis

Since RT and HR showed a normal distribution (Shapiro-Wilks for relevant modality and congruence condition), they were analyzed using repeated measures analyses of variance (RM-ANOVA) with the attention-related sensory modality (visual and auditory) and congruence (congruent and incongruent) as within-subject factors. The Bonferroni test was used for paired post hoc comparisons.

For ERP analysis, we followed a two-step approach. First, we assessed the congruence effect within each attention modality by performing a cluster-based nonparametric randomization test (Maris and Oostenveld 2007), comparing congruent and incongruent trials. Second, to examine whether the locus of attention modulated the congruence effect, we compared the differential ERP (i.e., the ERPce: congruent minus incongruent) between visual and auditory attention modalities using the same cluster-based test. The cluster-based test controls for the Type I error rate in situations involving multiple comparisons over sensors and times by clustering neighboring sensors (considering three adjacent channels) and time points that show the same effect. Given the exploratory nature of the study, for this analysis we included all channels and explored differences all along the epoch. Sensors for which the t value of the difference between conditions exceeds an a priori threshold ($p < 0.05$) were selected and subsequently clustered based on spatial adjacency, and the sum of the t-values within a cluster was used as a cluster level statistic. The cluster with the maximum sum was used as a test statistic. By randomly permuting the data across the two conditions and recalculating the test statistic 500 times, we obtained a reference distribution to evaluate the statistical significance of a given effect (Monte Carlo estimation). The null hypothesis distribution was generated empirically without assumptions about the data distribution.

In addition, we examined the relationship between the ERPce amplitude and behavioral performance (RTce and HRce) using another cluster-based permutation correlation analysis (Maris and Oostenveld 2007). Specifically, we tested whether individual differences in behavioral congruence effects were correlated with ERPce amplitude across sensor–time pairs. To this end, we used a nonparametric randomization procedure with 500 permutations for each attention modality, where the empirical correlation coefficients (Pearson's r) at each channel and time point were compared against a null distribution generated by randomly shuffling the behavioral values across participants. Sensors and time points exceeding an initial threshold ($p < 0.05$) were clustered based on spatial and temporal adjacency, and the sum of the correlation coefficients within a cluster was used as the cluster-level statistic. The largest cluster statistic was then compared to the permutation distribution to determine significance, thus controlling for the multiple comparison problem across electrodes and time. All results were considered statistically significant when the p value was below 0.05.

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3 | Results

3.1 | Behavior

The RT analysis exhibited a main effect of the congruence factor, with faster responses for the congruent compared to the incongruent trials ($F(1,29) = 76.96$, $p < 0.001$, $n_p^2 = 0.72$, mean difference (MDiff) = −96.82, Figure 2A). A significant main effect of modality was observed, with faster responses for visual attention compared to auditory attention ($F(1,29) = 62.17$, $p < 0.001$, $n_p^2 = 0.68$, MDiff = −110.18, Figure 2B). A significant congruence by modality interaction was also found ($F(1,29) = 6.94$, $p = 0.01$, $n_p^2 = 0.19$, MDiff = −65.36). Post hoc comparisons showed a stronger reduction in RT for the congruent versus incongruent trials in visual attention compared to auditory attention (MDiff = −129.50, Cohen's $d = -0.74$, and MDiff = −64.14, Cohen's $d = -0.36$, respectively; Figure 2C).

The HR analysis showed a main effect of the congruence factor, with higher HR for the congruent compared to the incongruent trials ($F(1,29) = 90.69$, $p < 0.001$, $n_p^2 = 0.75$, MDiff = 6.87, Figure 3A). Also, there was a significant main effect of modality, with higher HR for visual compared to auditory attention ($F(1,29) = 4.31$, $p = 0.04$, $n_p^2 = 0.13$, MDiff = 1.79, Figure 3B). A trend for an interaction of congruence by modality was also found ($F(1,29) = 3.70$, $p = 0.06$, $n_p^2 = 0.11$). Post hoc comparisons showed a higher increase in HR for the congruent versus incongruent trials in visual attention compared to auditory attention (MDiff = 8.22, Cohen's $d = 0.63$, and MDiff = 5.07, Cohen's $d = 0.42$, respectively; Figure 3C).

3.2 | ERPs

For visual attention, we found a differential pattern between congruent and incongruent trials. Congruent trials evoked a significant reduction in an occipital positivity (P50) between 40 and 64 ms after stimuli onset (cluster statistic (CS) = −1158.57, $p = 0.008$, Cohen's $d = -0.64$), followed by an increased left-central negativity (N1) between 76 and 128 ms (CS = −1158.57, $p = 0.008$, Cohen's $d = -0.4804$). Subsequently, a later positive deflection (P2) was observed between 180 and 220 ms over fronto-central regions (CS = 50.54, $p = 0.01$, Cohen's $d = 0.45$). Finally, a positive deflection (P3) was observed between 228 and 380 ms over left regions (CS = 62.93, $p = 0.004$, Cohen's d = 0.51) (Figure 4A). In the auditory attention condition, the temporal profile of ERPs showed N1 and P2 components similar to those observed under visual attention. However, no statistically significant differences were found between congruent and incongruent trials across the ERP time course (all $p > 0.05$; Figure 4A). Nevertheless, we observed a trend toward a greater left-central positivity (P3) between 236 and 320 ms (CS = 710.39, $p = 0.10$, Cohen's $d = 0.45$, Figure 4B). The raw grand averages of ERPs for congruent and incongruent trials in each modality are shown in the Supporting Information.

As mentioned before, we isolated the congruence effect by subtracting the ERPs of incongruent trials from those of congruent trials in each sensory modality (ERPce). Visual inspection of the resulting ERPce waveforms (Figure 4B) revealed a temporal misalignment of components across modalities,

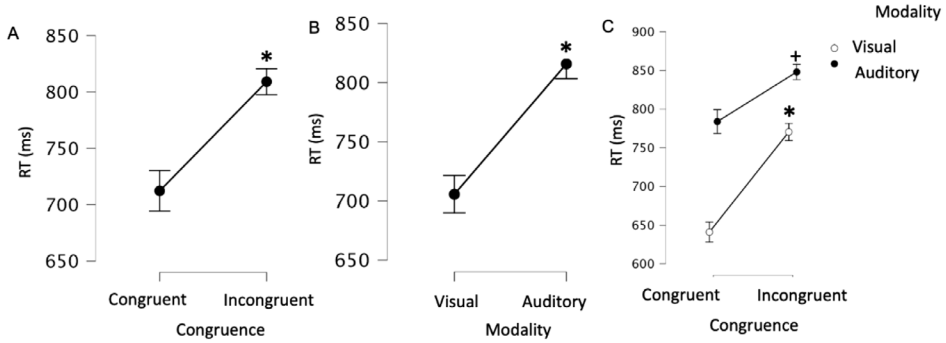


FIGURE 2 | Behavioral performance (RT). (A) A significant reduction in reaction times (RT) for the congruent compared to the incongruent trials was observed. (B) A significant reduction in RT for the visual compared to the auditory modality condition was observed. (C) Stronger reduction in RT for the congruent versus incongruent trials in visual attention compared to auditory attention. Error bars represent the standard error of the mean (SEM). *$p < 0.001$, +$0.001 < p < 0.05$.

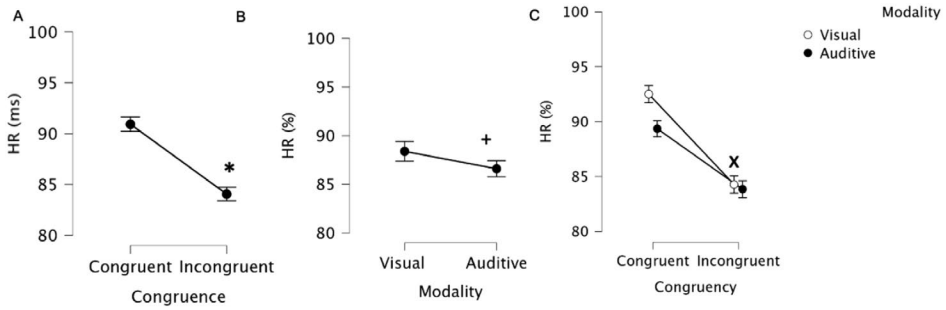


FIGURE 3 | Behavioral performance (HR). (A) A significant increase in accuracy was observed for the congruent condition compared to the incongruent condition. (B) Accuracy was also significantly higher for the visual condition compared to the auditory condition. (C) There was a trend toward an interaction between congruence and modality, with a greater increase in accuracy for congruent versus incongruent trials in visual attention compared to auditory attention. Error bars represent the standard error of the mean (SEM). *$p < 0.001$, +$0.001 < p < 0.05$, X $0.05 < p < 0.1$.

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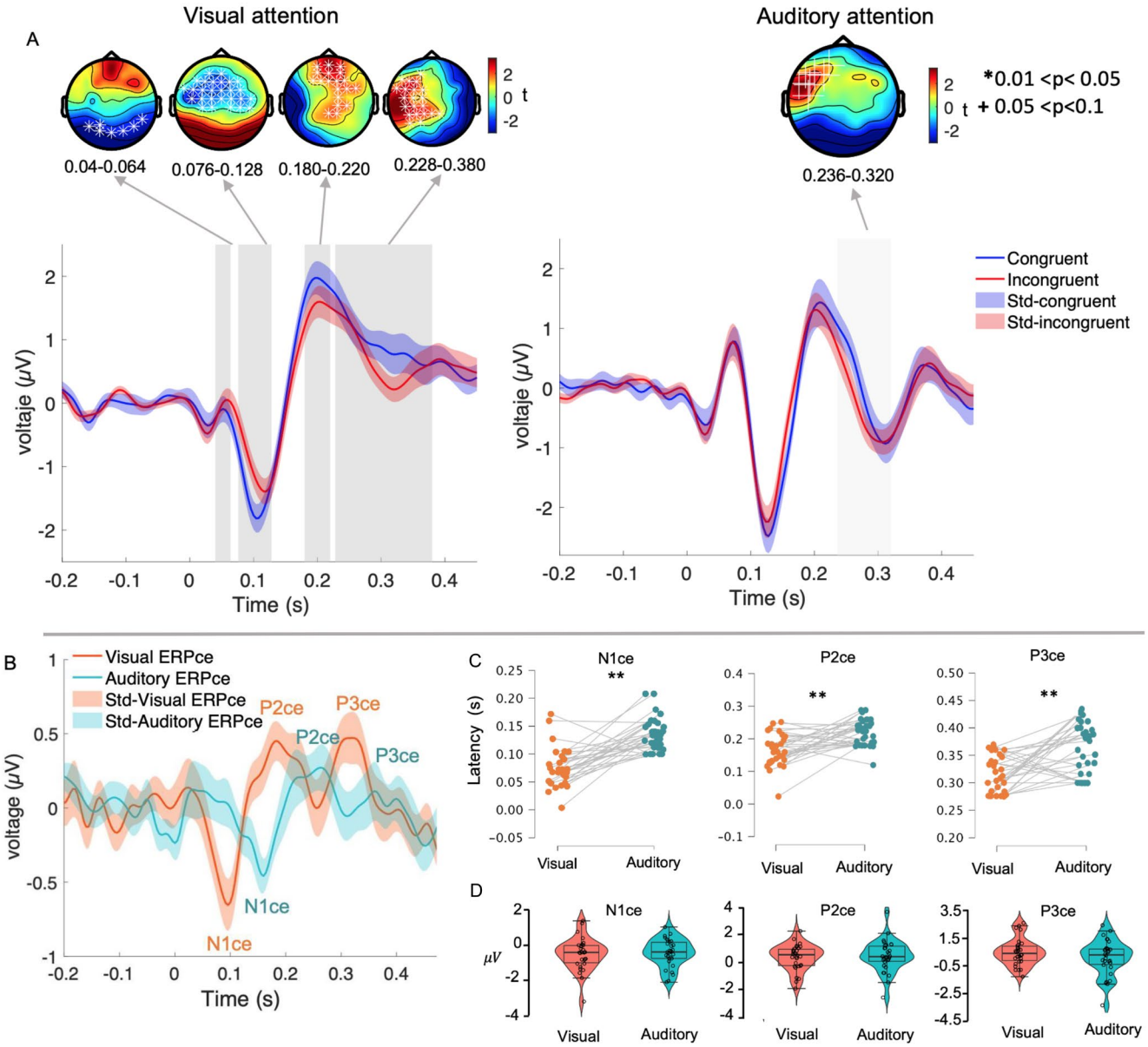


FIGURE 4 | Neurophysiological correlates of congruence for visual and auditory attention. (A) Grand average ERPs for congruent (blue) and incongruent (red) trials across occipital, temporal, and central channels (F7, F5, F1, F4, F6, FC5, FC3, FC1, FCz, FC2, FC4, FC6, C5, C3, C1, C2, C4, C6, CP5, CP3, CP1, CPz, CP2, CP4, CP6, P5, P3, P1, Pz, P2, P4, P6, O1, O2, Oz). The blue and red shadows correspond to the standard errors of the mean. The gray shading indicates temporal windows with significant differences (*$0.01 < p < 0.05$, +$0.05 < p < 0.1$). The topo-plots show the spatial distribution of the differences (T value of congruent versus incongruent trials) in each time window, and white asterisks indicate the channels involved in significant clusters ($p < 0.001$). (B) Neurophysiological correlates of the congruence effect (ERPce) (ERP congruent—ERPs incongruent) for both modalities (visual (orange) and auditory (turquoise)) across occipital, temporal, and central channels (the same as in A). The shadows correspond to the standard errors. (C) ERPce latencies for visual and auditory attention. **$p < 0.001$ (D) Mean ERPce amplitudes (N1ce, P2ce, and P3ce) for visual and auditory modalities, showing no significant differences between modalities ($p = 0.78$, $p = 0.71$, and $p = 0.20$, respectively). Violin plots display the probability density of the amplitude distribution, and box plots indicate the median and interquartile range (Q1–Q3).

particularly a delay in the onset of the N1ce and P2ce under auditory attention compared to visual attention. This observation prompted a quantification of component latencies using paired-sample t-tests. Latency was estimated using the fractional peak latency (FPL) method, defined as the time point at which the ERPce waveform reached 50% of its peak amplitude. Specifically, the N1ce latency was significantly longer during auditory attention (136 ± 30 ms, range: 100–208 ms) than during visual attention (75 ± 35 ms, range: 40–100 ms, $t(30) = 7.1$, $p < 0.001$, Cohen's $d = -1.28$). The P2ce followed the same pattern (auditory: 219 ± 40 ms, range: 120–288 ms; visual: 167 ± 50 ms, range: 24–252 ms, $t(30) = 4.2$, $p < 0.001$, Cohen's $d = -0.75$). The P3ce component also occurred significantly earlier in the visual condition (316 ± 34 ms, range: 276–368 ms) than in the auditory condition (365 ± 47 ms, range: 300–432 ms, $t(30) = -3.94$, $p < 0.001$, Cohen's $d = -0.74$; Figure 4C). To allow a fairer comparison of the temporal dynamics across modalities in the second cluster-based

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permutation analysis (comparison of the ERPce waveforms between modalities), we applied a temporal alignment to the auditory ERPce. This was done by shifting the auditory waveform by 61 ms (i.e., the average latency difference of the N1ce component between modalities). This adjustment was implemented prior to running the statistical comparison and aimed to minimize bias introduced by component latency differences. The alignment procedure was performed uniformly across participants and conditions and did not involve any trial-level modifications. However, no statistically significant differences were found between ERPce waveforms between modalities (all $p > 0.05$). Regarding amplitude, the mean N1ce amplitude was −0.83 μV (SD = 1.32 μV; range: −4.28 to 1.64 μV) during visual attention and −0.33 μV (SD = 0.82 μV; range: −2.50 to 1.20 μV) during auditory attention. For the P2ce component, the mean amplitude was 0.36 μV (SD = 0.93 μV) in the visual condition and 0.43 μV (SD = 1.09 μV) in the auditory condition. The P3ce component exhibited a mean amplitude of 0.49 μV (SD = 0.95 μV; range: −1.27 to 2.62 μV) in the visual modality and 0.12 μV (SD = 1.25 μV; range: −3.34 to 2.45 μV) in the auditory modality (Figure 4D). Notably, the P3ce component under auditory attention showed high inter-individual variability and low consistency, so conclusions regarding attentional effects on its latency in this condition should be interpreted with caution (Figure 4D).

3.3 | Association Between ERPs and Behavior

We explored the correlation strength between ERPce amplitude and RTce (congruent—incongruent). For visual attention, we observed a positive correlation between ERPce and RTce ($r = 0.38$, uncorrected $p = 0.02$, Fisher's $z = 0.39$) between 0.190 and 0.210 s over fronto-central electrodes (AF4, F3, F4, FT7, FC5, FC4, FC6, AF3, C5, C3, C6, CP3, CP1, CP2, CP4), coincident with the P2ce (Figure 5A). In the auditory attention condition, 5 time windows were identified in which brain activity significantly correlated with reaction times (RTs). These correlations were observed specifically at electrodes Fp1 and FC5, and occurred within the following time intervals: (1) 0.060–0.072 s: $r = 0.37$, uncorrected $p = 0.03$, Fisher's $z = 4.28$; (2) 0.096–0.124 s: $r = 0.34$, uncorrected $p = 0.04$, Fisher's $z = 3.53$; (3) 0.164–0.176 s: $r = 0.38$, uncorrected $p = 0.02$, Fisher's $z = 4.55$; (4) 0.212–0.216 s: uncorrected $r = 0.38$, $p = 0.02$, Fisher's $z = 4.55$; (5) 0.252–0.264 s: $r = 0.38$, uncorrected $p = 0.02$, Fisher's $z = 4.55$, coincident with the N1ce and P2ce (Figure 5B). Moreover, in the visual attention condition, we also observed a positive correlation between ERPce and HRce ($r = 0.48$, uncorrected $p = 0.008$, Fisher's $z = 0.52$; Figure 5C) from 0.164 to 0.180 s over centro-parietal electrodes (F1, FC1, FC2, FC6, Cz, C4, CP1, CP2, CP6, F2, Pz, P1, P2, PO3, POz), corresponding to the early portion of the P2ce. For auditory attention, ERPce and HRce were also positively correlated ($r = 0.53$, uncorrected $p = 0.003$, Fisher's $z = 0.59$; Figure 5D) between 0.172 and 0.192 s over fronto-central channels (FC6, CP5, CP3, CP1, CP4, CP6, P5, P3, P1, PZ, P2, POz, Oz, O2).

It is important to note that, due to the polarity of the ERP components, positive correlations must be interpreted in context. For the N1ce (a negative-going component), a positive correlation with RTce implies that more negative amplitudes are associated with faster RT in congruent versus incongruent conditions during auditory attention. In contrast, for the P2ce (a positive-going component), a positive correlation indicates that larger amplitudes are associated with slower RT and higher HR under both visual and auditory attention. Therefore, while the correlation coefficients are positive in all cases, their functional interpretation differs depending on the ERP polarity. However, none of these correlations survived correction for multiple comparisons, and thus they must be interpreted with caution.

3.4 | Association of ERPs Latencies Between Modalities and Behavioral Performance

To further explore the role of the timing of ERPce components during AV integration, we examined the association between ERPce latency and RTce separately for each modality. No significant correlations were found for either the visual or auditory condition. In the visual attention condition, the correlations were as follows: N1ce: $r = -0.10$, $p = 0.58$, Fisher's $z = -0.10$; P2ce: $r = 0.08$, $p = 0.65$, Fisher's $z = 0.08$; and P3ce: $r = 0.04$, $p = 0.82$, Fisher's $z = 0.04$. In the auditory attention condition, the results were: N1ce: $r = 0.10$, $p = 0.58$, Fisher's $z = 0.10$; P2ce: $r = -0.03$, $p = 0.86$, Fisher's $z = -0.03$; and P3ce: $r = 0.08$, $p = 0.66$, Fisher's $z = 0.08$.

4 | Discussion

In this study, we examined whether the attentional locus affects the neurophysiological characteristics of the congruence effect between synchronous AV stimuli within a crossmodal attention task, particularly under conditions with a high proportion of incongruent trials. We describe an ERP waveform related to the congruence effect (ERPce) consisting of an N1-P2-P3 complex with a left fronto-central distribution, regardless of the attended sensory modality, which was associated with behavioral performance. Our findings indicate that the attentional locus significantly influences the temporality of the congruence effect, as revealed by delayed latencies of the ERPce for auditory versus visual attention. These results suggest that the attentional locus plays a significant role in integrating simultaneous congruent (though irrelevant) signals with relevant ones in a context of broad incongruency. This integration appears to be mediated by the N1ce and P2ce components over high-level association brain regions.

4.1 | Behavioral Findings

Our results revealed that participants exhibited improved performance in congruent conditions compared to incongruent ones, as indicated by RT and HR in both visual and auditory attention. These findings are consistent with the congruence effect as reported by numerous studies (Mastroberardino et al. 2015; Mishra and Gazzaley 2012; Misselhorn et al. 2016; Solis-Vivanco et al. 2018). Interestingly, we found that this congruence effect occurred despite frequent irrelevant distractors. This reflects that, even when inhibiting distractors was useful most of the

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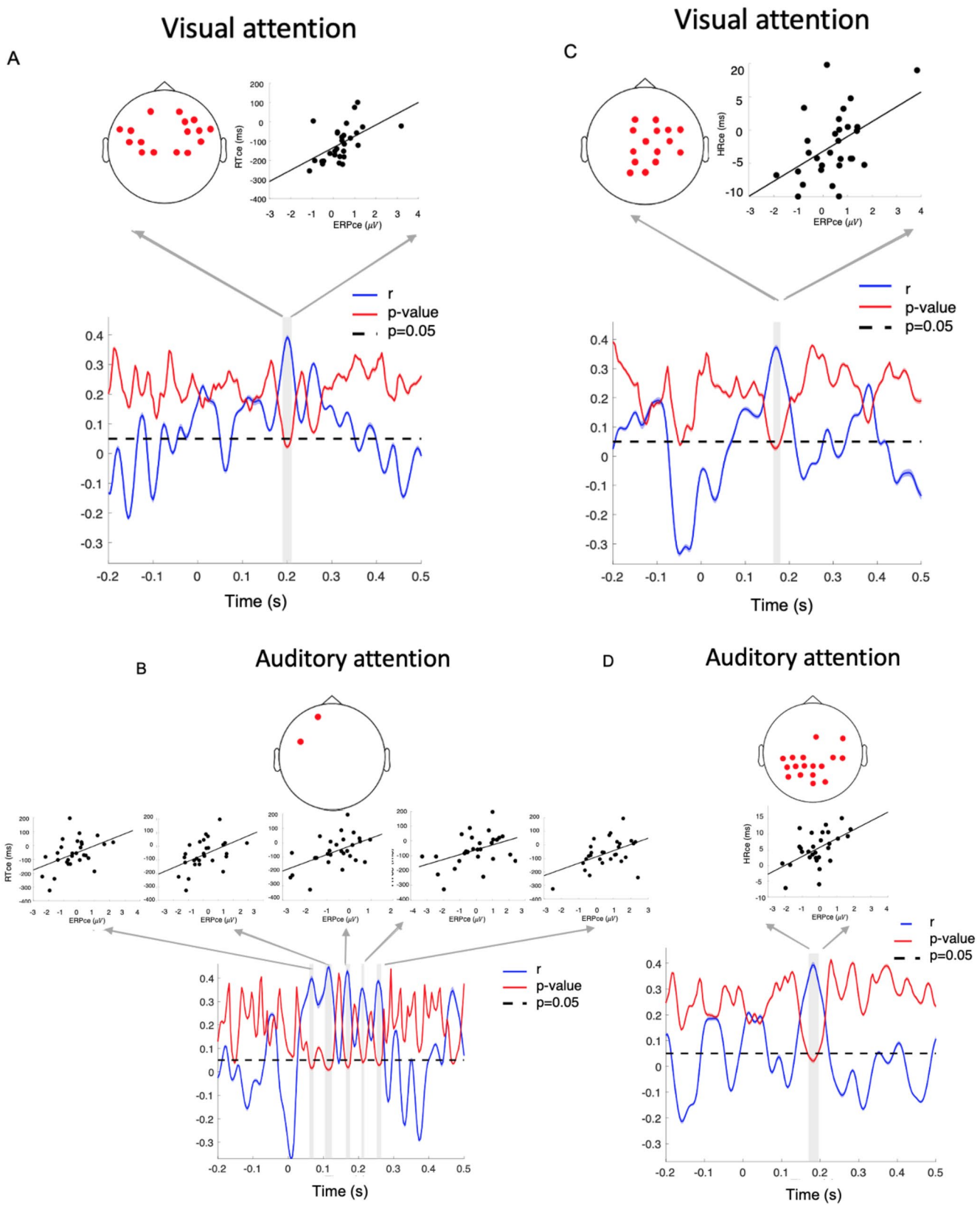


FIGURE 5 | Legend on next page.

time, when the irrelevant stimuli were congruent, the system allowed them to be processed alongside relevant ones, potentially enhancing the perceptual processing of both (Lowe and Mitterer 1982; Schmidt et al. 2014; Talsma et al. 2010).

The observed larger congruence effect during visual attention suggests that auditory information—although task-irrelevant—enhances performance when it is congruent with the attended visual input. We interpret this pattern as evidence that the

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FIGURE 5 | Spatiotemporal correlations between ERP congruence effects (ERPce) and behavioral measures (RTce and HRce) during visual and auditory attention. Correlations were computed across all electrodes and within a −200 to 1100 ms window relative to stimulus onset, generating spatiotemporal brain–behavior association maps. At each time point, ERPce amplitudes were correlated with behavioral performance across participants. (A) During visual attention, the ERPce amplitude positively correlated with RTce between 190 and 210 ms over fronto-central electrodes: AF4, F3, F4, FT7, FC5, FC4, FC6, AF3, C5, C3, C6, CP3, CP1, CP2, CP4. (B) In the auditory attention condition, 5 time windows showed significant positive correlations between ERPce amplitude and RTce, which correspond to the N1ce and P2ce time range. These were observed at electrodes Fp1 and FC5. (C) For visual attention, ERPce amplitude positively correlated with HRce corresponding to P2 over fronto-central electrodes: F1, FC1, FC2, FC6, Cz, C4, CP1, CP2, CP6, F2, Pz, P1, P2, PO3, and POz. (D) During auditory attention, ERPce amplitude was positively correlated with HRce corresponding to P2 over fronto-central electrodes: FC6, CP5, CP3, CP1, CP4, CP6, P5, P3, P1, Pz, P2, POz, Oz, and O2. None of the correlations survived correction for multiple comparisons.

attentional locus modulates the behavioral congruence effect, such that during visual attention, there is facilitated detection and integration of congruent signals from the unattended (auditory) modality, operating in parallel with visual processing. This interpretation is consistent with our ERPce results, which indicate enhanced neural activity associated with multisensory integration during visual attention and aligns with prior findings showing modality-dependent cross-sensory interactions (Donohue et al. 2013).

4.2 | Neurophysiological Processing of Congruent and Incongruent Information for Visual and Auditory Attention

For visual attention, the analysis of the neurophysiological processing of congruent versus incongruent information revealed an early decrease of a positive ERP in occipital regions between 40 and 60 ms (P50). This reduction in neural activity, possibly indicative of sensory gating mechanisms, supports the notion that auditory stimulation influences visual analysis even before it is completed in the occipital cortex during early sensory-perceptual processing (Fort et al. 2002; Giard and Peronnet 1999; Mishra and Gazzaley 2012; Molholm et al. 2004; Raij et al. 2010). It is important to note that the congruence of our AV stimuli could be detected as early as 40 ms post-onset, when the vowel (which indicated the congruence or incongruence) could be detected. This finding aligns with previous research on AV integration and crossmodal attention, revealing that when the onset of auditory congruent information precedes the onset of the visual one, a decrease in the activity of visual regions can be found (Boutros et al. 2013; Freedman et al. 1987; Yadon et al. 2009). This phenomenon may be attributed to the brain's ability to "gate out" redundant stimuli, that is, when the presentation of congruent auditory distractors influences early visual processing, resulting in a decreased neural response to relevant visual stimuli, mainly reflected by a reduction of P50 responses (Anokhin et al. 2007).

In addition, the increased negativity over fronto-central regions of the N1 (70–128 ms) for congruent trials during visual attention indicates enhanced processing of auditory distractors (Paris et al. 2017; Vroomen and Stekelenburg 2010). This finding suggests a possible involuntary capture of attention (Solís-Vivanco et al. 2021) and indicates that even under frequent incongruence, the gating and integration of auditory information remains possible. The P3 component observed over left temporal regions from 280 to 340 ms and over occipital regions from 348 to 380 ms might reflect a feedforward of auditory input reaching areas of multisensory convergence, such as the superior temporal region, and being transmitted via feedback connections to earlier visual areas (Molholm et al. 2002). Together with the observed early occipital P50, this ERP waveform might reflect the flow of cortical processing information from irrelevant primary sensory to associative and subsequent feedback to relevant sensory areas as a consequence of the presentation of congruent and simultaneous AV stimuli (Koelewijn et al. 2010).

For auditory attention, the ERPs waveforms (P50, N1, P2, and P3) were observed for both congruent and incongruent bimodal trials. However, no significant differences were detected for any component. The similar activity between congruent and incongruent trials for P50 suggests that the visual distractors were processed regardless of their congruence. Regarding the auditory N1 and P2, the absence of differences between congruent and incongruent AV vowels reflects that the impact of visual distractors on the processing of auditory relevant stimuli was similar. This aligns with well-documented findings showing that auditory signals typically reach the cortex within 10–15 ms due to mechanical signal transduction in the cochlea. Visual signals are typically associated with longer latencies compared to auditory ones, with activation in the occipital cortex (V1) reported to begin around 50–60 ms after stimulus onset, reflecting the cumulative effects of retinal transduction and signal transmission (Foxe and Simpson 2002; Wilson et al. 1983). Consequently, in the context of cortical visual pathways, visual stimulation may begin to impact auditory processing after approximately 45–50 ms, once early visual cortical regions have been engaged and integration can occur at higher-order multimodal sites. Therefore, temporal differences in the arrival of visual and auditory stimuli to the cortex suggest that visual congruent distractors would enhance relevant auditory information at a later stage. This might explain the lack of significant differences in the ERPs between congruent and incongruent trials for auditory attention during early time windows.

Overall, our results imply that the timing of the integration between irrelevant congruent and relevant stimuli is heavily influenced by attentional locus, even in the presence of considerable incongruent distractors.

4.3 | ERP of the Congruence Effect for Visual and Auditory Attention

To further elucidate the role of attentional locus on the neurophysiological processing of the congruence effect, we extracted

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the ERPce (ERP congruent—ERP incongruent) in each modality, which allowed us to eliminate the effect of sensory activity within and between modalities. When comparing the ERPce between visual and auditory attention, we observed a temporal lag of approximately 50 ms for the N1ce and P2ce components. As mentioned before, visual signals are typically associated with longer latencies (Foxe and Simpson 2002; Wilson et al. 1983). The absence of significant congruence effects in auditory ERPs during auditory attention may reflect a limited influence of visual inputs at early stages of auditory processing, potentially due to temporal asynchronies in the arrival of cross-modal information to higher-order integrative regions. Additionally, this asymmetry may also reflect differences in attentional prioritization and inhibitory control across modalities. Previous studies have shown that visual information tends to dominate in multisensory contexts, making it more difficult to suppress when it is irrelevant during auditory attention (Matusz et al. 2017; Talsma et al. 2010). This could help explain why irrelevant visual stimuli may interfere more strongly during auditory attention than irrelevant auditory stimuli during visual attention.

Importantly, we confirmed that the ERPce components (N1ce, P2ce, and P3ce) from both modalities exhibited similar voltages and fronto-central distributions. This finding indicates that, although attentional allocation influences the timing of AV information integration, particularly for the N1ce and P2ce components, such integration occurs with similar features at the high-level association cortex regardless of attentional focus on either modality.

4.4 | Association Between Neurophysiology and Behavior

Although the correlations between ERPce and behavioral congruence effects (RTce and HRce) did not survive correction for multiple comparisons, the uncorrected findings suggest potentially meaningful associations that warrant further investigation. Specifically, for visual attention, larger P2ce amplitudes were associated with higher HRce, indicating that participants were better able to capitalize on sensory congruence to enhance response accuracy. Simultaneously, higher P2ce amplitudes were correlated with smaller RTce values (e.g., lower congruence effects). We consider that these results are not contradictory but rather may reflect distinct yet complementary facets of multisensory processing: while enhanced HRce captures participants' capacity to exploit sensory congruence for improved performance accuracy, reduced RTce may reflect less sensitivity to congruence differences between stimuli. A larger P2ce, therefore, may index a more robust neural response that supports both enhanced precision and reduced effect of conflict in response speed. This interpretation aligns with prior evidence linking P2 modulations to the detection and resolution of AV incongruence, particularly at the phonetic level (Stekelenburg and Vroomen 2007). Therefore, here we propose the P2ce component, across both visual and auditory attention, as a candidate neural marker associated with individual differences in the ability to manage multisensory conflict.

In addition to P2ce, exploratory correlations also implicated the N1ce component, particularly under auditory attention, as a potential neural correlate of multisensory processing. Although the observed positive correlation between N1ce amplitude and RTce did not survive correction for multiple comparisons, it revealed that more negative N1ce responses were associated with faster RT in congruent compared to incongruent trials (i.e., stronger congruence effect). This finding is in line with previous evidence suggesting that the N1 component plays a key role in the integration of temporally congruent AV signals (Paris et al. 2017; Vroomen and Stekelenburg 2010). Given the known sensitivity of the N1 to attentional and perceptual processing (Ho et al. 2015), these findings suggest that attentional mechanisms may contribute to the speed with which individuals respond to temporal congruent multisensory information, particularly under auditory attention.

Taken together, these exploratory findings point to the N1ce and P2ce components as potential neural signatures of adaptive multisensory integration and attentional control. Future studies should aim to replicate these associations using larger sample sizes and corrected statistical thresholds, and further elucidate the distinct functional contributions of both N1ce and P2ce in tasks involving AV speech perception and selective attention.

4.5 | Association Between ERPs Latencies and Behavioral Performance

In the present study, we examined whether the latency of ERPce components within each modality was associated with the magnitude of behavioral congruence effects (RTce). The rationale was that shorter ERPce latencies, particularly in early components (e.g., N1ce, P2ce), could reflect more efficient temporal processing and, in turn, support faster or more consistent behavioral responses to multisensory congruent events. However, no significant correlations were observed between ERPce latency and RTce in either the visual or the auditory attention condition. These results suggest that the timing of ERPce components may not be a strong determinant of behavioral congruence effects. It is also possible that the relationship between neural latency and behavioral performance is more complex than a simple linear association, or that it emerges only under specific task demands or in particular populations.

The absence of statistically significant associations, together with relatively small effect sizes, underscores the exploratory nature of this analysis. Future studies with larger sample sizes and finer-grained measures of neural timing may help to better elucidate its potential role in shaping behavioral outcomes during multisensory integration. Furthermore, the influence of top-down attentional factors on ERP–behavior relationships remains an open question, particularly in light of the modality-specific differences observed in RTce.

4.6 | Implications for Multisensory Integration

This study demonstrates that the attentional locus influences the timing of neurophysiological processes involved in multisensory congruent information, as revealed by the N1ce and P2ce components. This modulation enhances the perception of relevant information even under highly incongruent contexts.

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Our findings suggest that despite the increased attentional control demands due to a high proportion of irrelevant distractors, the congruence effect of simultaneous AV information persists. This effect likely occurs in higher-level association cortex regions (left temporal–fronto–central areas) and potentially serves as a predictor of behavioral performance.

4.7 | Limitations of This Study

Although our study provides strong evidence that attention promotes similar neurophysiological processing of congruent distractors regardless of its locus during simultaneous presentation of AV stimuli in a crossmodal attention task, there are limitations to consider. First, we did not control for the timing of the arrival of the stimuli at the cortex. To confirm the role of N1-P2 components on the integration of simultaneous congruent signals, future studies should adapt the stimulus-onset asynchrony (SOA) between visual and auditory vowels, considering the arrival time of each unisensory information at the cortex. Another limitation of this study is that we did not conduct source localization analyses. Such analysis could help corroborate whether the left temporal regions implicated in our findings of the congruence effect across modalities are indeed the left superior temporal cortex and left temporal pole, which have been identified as a central node in AV integration (Gao et al. 2023; Ohki et al. 2016). However, this was not part of the objectives of the study.

5 | Conclusion

In the present study, we found that the congruence effect in both visual and auditory attention is manifested by the N1ce and P2ce ERPs at an early processing stage (approximately 100–200 ms) and predominantly in the same left fronto-central regions. However, we observed that the attentional focus has an impact on the temporal dynamics of the congruence effect, as revealed by a delay in the auditory compared to visual attention ERPs. This suggests that despite the capacity of auditory information to reach the cortex faster, the integration of visual information requiring additional processing time may impact the timing efficiency of multisensory integration. Our ERP results, combined with behavioral outcomes, suggest that the congruence effect, even in contexts with a high proportion of incongruent distractors, may be modulated by the locus of attention. This finding highlights a novel aspect of temporal processing during multisensory integration, providing insights into attentional biases in real-world multisensory environments.

Author Contributions

Edwin R. Ramírez-Benítez: conceptualization, data curation, formal analysis, investigation, methodology, validation, writing – original draft, writing – review and editing. **Alejandra E. Ruiz-Contreras:** conceptualization, supervision, validation, writing – original draft, writing – review and editing. **Alejandra Mondragón-Maya:** conceptualization, supervision, validation, writing – original draft, writing – review and editing. **Rodolfo Solís-Vivanco:** conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, supervision, writing – original draft, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data utilized in this study and analytic code can be shared upon reasonable request to the authors.

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Supporting Information

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